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# A Multi-Agent Generative AI Framework for Supporting Beyond Standard Model Physics Research



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## Abstract

# 1 Introduction

The advancement of technology and progress in physics research have been deeply intertwined. Fundamental research in physics has historically driven technological innovation, which in turn has accelerated the pace of discovery by enabling faster data acquisition, processing, and analysis. This relationship is perhaps most clearly demonstrated in the experiments at the Large Hadron Collider (LHC), particularly in the ATLAS and CMS experiments [1, 2]. Each experiment is equipped with its own dedicated detector, designed to record proton-proton collisions at an extraordinary rate—producing vast volumes of data per second. To conduct these experiments with greater precision and efficiency, continuous improvements in both detector technology and computational tools have been essential. While detector development focuses on improving sensitivity, resolution, and readout speed, advances in computing increasingly emphasize data processing, simulation, and, more recently, machine learning-driven analysis. Machine learning has been employed in high-energy physics since its early use in charged particle tracking with neural networks [3], and played a crucial role in the discovery of the Higgs boson by enhancing the sensitivity to its decay into  $\tau$  leptons using boosted decision trees [4], illustrating its growing importance in LHC analyses [5]. As computing capabilities continued to advance, more sophisticated methods such as deep learning became viable, offering significant advantages for large and complex datasets. These developments have directly benefited high-energy physics experiments. Deep learning techniques have demonstrated strong performance in tasks such as particle identification, jet substructure analysis, and event classification, particularly in data-intensive scenarios [6]. Further advancements in deep learning have led to the development of large language models (LLMs), which have had a revolutionary impact across industry, academia, and a wide range of disciplines. These state-of-the-art AI systems are capable of understanding and generating human language. While earlier AI developments primarily focused on data-driven tasks such as classification or regression, LLMs introduce new capabilities including scientific reasoning, summarization, interactive document querying, idea generation support, and fine-tuned task adaptation—broadening their potential to assist in complex research workflows. However, general-purpose LLMs often fall short when applied to physics, where symbolic reasoning, domain-specific terminology, and methodological rigor demand models that are specialized and aligned with disciplinary standards [7]. To address this challenge, new advancements and more sophisticated model architectures are required. One such approach is Retrieval-Augmented Generation (RAG), a framework that enhances the reasoning and factual grounding capabilities of language models by retrieving relevant external information and incorporating it at inference time [8]. For more complex architectures, multiple large language models can be specialized for particular tasks through techniques such as task-specific prompting, connecting to dedicated databases, or applying tailored reasoning strategies. These specialized models are often referred to as agents. By coordinating the behavior of such agents, multi-agent systems can be constructed that outperform single-model setups, enabling more robust, modular, and context-aware reasoning pipelines. In this study, a multi-agent framework is developed to reason over physical context, interpret domain-specific terminology, read and analyze tables and figures containing numerical data, link information to relevant textual content, and generate meaningful answers to user queries. To achieve this, distinct agents are designed to handle reasoning, database construction, and retrieval tasks. More specifically, the framework is tailored for high-energy physics and is applied to Supersymmetry

(SUSY) publications from the CMS Collaboration. The goal is to build a system that is aware of the literature produced by CMS in the context of Beyond the Standard Model (BSM) searches, and that can interpret, analyze, and answer questions related to the current state of research in this domain.

## 2 Related Work

## 3 Physics Beyond the Standard Model

## 4 System Architecture

## 5 Retrieval and Reasoning Methodologies

## 6 Results

## 7 Conclusion

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